Collaborate and Listen: International Research Collaboration at Learning @ Scale

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ABSTRACT

International research collaboration (IRC) leads to stronger, more visible research, and it allows researchers to account for global perspectives in their work. This is particularly important in Learning @ Scale research, as "scale" often implies learning that spans countries or continents. This paper examines IRC at Learning @ Scale by analyzing the authorship of all Learning @ Scale papers over the conference's 10-year history. We find that IRC at the conference is low, and possibly trending downwards. Additionally, authors from South American, African, and Asian countries tend to favor IRC, and those countries that publish most frequently at Learning @ Scale tend to be less internationally collaborative. This paper serves as a call to action for researchers and the conference to increase IRC, and we provide recommendations for accomplishing this goal.

CCS CONCEPTS

• **Applied computing** → **Distance learning**; • **General and reference** → **Surveys and overviews**.

KEYWORDS

international research collaboration, IRC, review

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1 INTRODUCTION AND RELATED WORK

International research collaboration (IRC) offers multiple benefits to researchers, their institutions, and the field of study, including: increased visibility and citation of research; distribution of costs and resources; greater creativity; and overall stronger research [5, 11]. Further, Kim notes that, in addition to promoting and facilitating knowledge exchange, "International collaboration allows researchers to take into consideration the impact of that research in a different culture" [4]. This observation is particularly relevant to the field of at-scale learning, where the concept of expanding availability of and access to education implies the need for a global perspective, which may be more easily attained through IRC than through solely domestic collaboration (or no collaboration at all).

This work-in-progress paper examines IRC at the Learning @ Scale conference since its inception in 2014. While several synthesis papers have been published at Learning @ Scale, these papers have focused largely on categorizing research topics rather than authorship [6–8, 10]. Only Duncan, et al, conducted an authorship analysis, which focused on other conferences where Learning @ Scale authors publish [2]. Building on this work by looking at authorship through the lens of international collaboration, this paper examines three research questions:

- **RQ1:** To what extent is IRC present at Learning @ Scale, both overall and over time?
- **RQ2:** Which countries and continents tend to be the most (and least) internationally collaborative?
- **RQ3:** Is there any correlation between the number of Learning @ Scale papers from a country, and the tendency for authors from that country to collaborate internationally?

2 METHODOLOGY

There have been 562 total papers published at Learning @ Scale from 2014 through 2023. A team of four coders—students and alumni in an online graduate computer science master's program—gathered authorship data about each paper. Each coder was assigned roughly 25% of the 562 papers. For each paper in their subset, each coder documented the objective details of the paper: title, year, classification (full paper or WIP), and session name (if applicable), as well as the names, affiliations, and countries for every author.

We focused our analysis of these data on the countries represented by the authors, and we assessed international collaboration from two perspectives: overall collaboration over time, and collaboration by country and continent. For the former, we first determined the number of countries that appeared on each paper, regardless of the number of authors on the paper. (For example, a paper with five authors all from the United States has one country represented among its authors; a paper with three authors—one from the United States and two from Germany—has two countries represented.) We used these numbers to calculate the average number of countries

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represented per paper for each year of Learning @ Scale, as well as the percent of papers each year that had at least two countries represented in their authorships (subsequently referred to as "collaborative papers"), in keeping with the definition of "international research collaboration" used elsewhere in the literature [1, 5, 9].

To understand collaboration by country and continent, we determined the number of collaborative papers in whose authorship each country and continent appear and divided this by the total number of papers in whose authorship that country or continent appears. Using this denominator (rather than the total number of Learning @ Scale papers) gave us a measure of the tendency for authors from a certain country or continent to collaborate internationally.

Additionally, we ordered each country by the number of Learning @ Scale papers it appears on, and we separately ordered each country by the percentage of its papers that are collaborative. Using these two lists, we examined the number of collaborative papers for those countries that most frequently appeared in Learning @ Scale authorships, as well as the reverse—the number of papers for those countries with the highest percentages of collaborative papers. This allowed us to understand whether papers from countries that frequently publish at Learning @ Scale tend to be collaborative or not, as well as whether countries that frequently appear on collaborative papers tend to appear on Learning @ Scale papers.

3 RESULTS

Our results are broken down into two general categories: IRC over time and IRC by country and continent.

3.1 International Research Collaboration Over Time

41 countries across six continents have appeared in the authorship of Learning @ Scale papers since its inception. The average Learning @ Scale paper contains one country in its authorship (rounded to the nearest whole number). This is true for each of the 10 years of the conference, as well as cumulatively across all 10 years.

Figure 1 shows percentages of collaborative papers over time. We found that 18% of all papers are collaborative. The percentage increased from 2014, when there were no collaborative papers, although it peaked at 31% in 2017 and has decreased every year since then except for 2021, when it rose to 26% from 10% in 2020—possibly facilitated by greater IRC afforded by virtual research in the wake of the COVID-19 pandemic.

3.2 International Research Collaboration by Country and Continent

The results of our analysis of collaboration at the country level are shown in Figure 2. We focused our analysis on countries that had at least three collaborative papers at the conference overall, since it was not meaningful to extrapolate and draw conclusions with fewer than three data points.

100% of papers whose authorship featured Denmark or Ecuador were collaborative, followed by Singapore (91.7%), Switzerland (85.7%), Brazil (83.3%), China (83.3%), India (81.8%), and Netherlands (81.8%). Notably, only 18% (the lowest of all countries) of papers whose authorship featured the United States were collaborative, in spite of the United States appearing on the vast majority

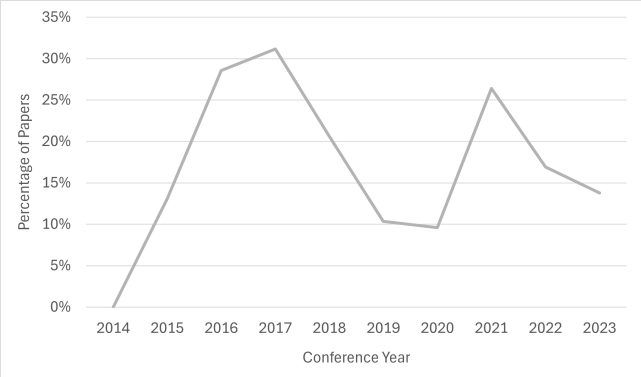

Figure 1: Percentage of collaborative Learning @ Scale papers over time.

(76.2%) of Learning @ Scale papers, and this was the only country with a percentage under 20%.

The results of this same analysis at the continent level are shown in Figure 3. For five of the six continents analyzed, at least 49% of the papers featuring authorship from those countries were collaborative, with South America claiming the highest percentage (90%). Only North America fell below this percentage—20.3% of papers featuring North American authors were collaborative papers.

Table 1 shows countries in the top 10 in terms of the percentage of all Learning @ Scale papers a country appears on, along with the percentage of each country's papers that are collaborative and the corresponding ordering based on that percentage. Table 2 shows countries in the top 10 in terms of the percentage of that country's papers that are collaborative, along with the percentage of all Learning @ Scale papers that country has appeared on and the corresponding ordering based on that percentage. (Note that each list contains more than 10 countries due to some countries being tied for a certain positions.) For each of the two metrics, there were 17 total ordered positions.

Only one of the top five countries (Canada) based on total number of Learning @ Scale papers appears in the top 10 countries with the highest percentages of collaborative papers (#8); in fact, nearly half of the countries in the top 10 based total number of papers are absent from the top 10 collaborative countries list. Notably, the United States appears on the most Learning @ Scale papers by far, but it is last based on its percentage of collaborative papers.

Four of the eight countries in the top five based on number of collaborative papers appear in the top 10 countries with the highest number of Learning @ Scale papers, and none of these four countries are above #8. Additionally, over half of the countries in the top 10 based on number of collaborative papers are absent from the top 10 list based on total Learning @ Scale papers.

No South American countries appear on the top 10 list for total number of papers, while two South American countries appear in the top four countries with the highest percentages of collaborative papers. No African countries appear on either list.

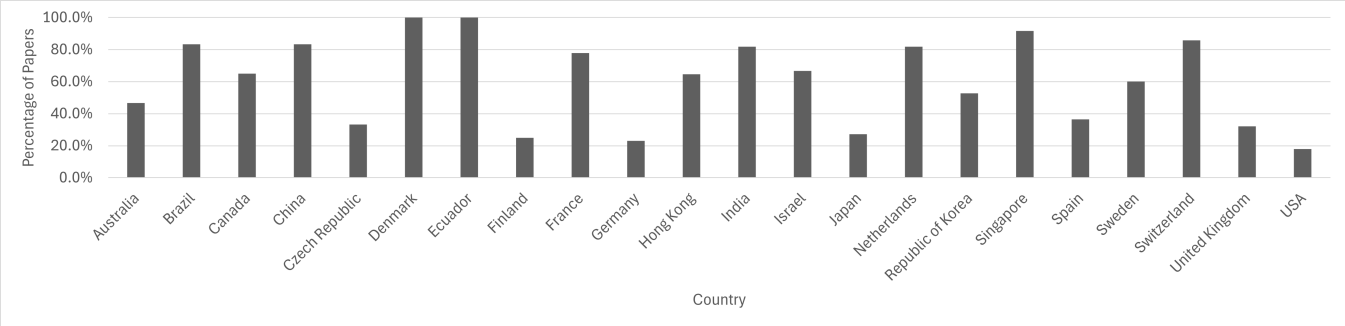

Figure 2: Percentage of collaborative Learning @ Scale papers by country.

Table 1: Top 10 ordering of countries based on the percentage of all Learning @ Scale papers for which each country appears in the authorship. Collaboration position is based on the percentage of a country's papers that are collaborative.

Position	% of L@S Papers Appeared On	Country	Continent	Collaboration Position	% Collaborative Papers
1	76.2%	United States	North America	17	18.0%
2	5.0%	United Kingdom	Europe	14	32.1%
3	4.6%	Germany	Europe	17	23.1%
4	3.6%	Canada	North America	8	65.0%
5	3.4%	Republic of Korea	Asia	11	52.6%
6	3.0%	Hong Kong	Asia	9	64.7%
7	2.7%	Australia	Oceania	12	46.7%
8	2.1%	China	Asia	4	83.3%
8	2.1%	Singapore	Asia	2	91.7%
9	2.0%	India	Asia	5	81.8%
9	2.0%	Japan	Asia	15	27.3%
9	2.0%	Netherlands	Europe	5	81.8%
9	2.0%	Spain	Europe	13	36.4%
10	1.6%	France	Europe	6	77.8%
10	1.6%	Israel	Asia	7	66.7%

Table 2: Top 10 ordering of countries based on the percentage of each country's papers that are collaborative. Total papers position is based on the percentage of all Learning @ Scale papers for which each country appears in the authorship.

Position	% Collaborative Papers	Country	Continent	Total Papers Position	% of L@S Papers Appeared On
1	100.0%	Denmark	Europe	15	0.5%
1	100.0%	Ecuador	South America	15	0.5%
2	91.7%	Singapore	Asia	8	2.1%
3	85.7%	Switzerland	Europe	11	1.2%
4	83.3%	Brazil	South America	12	1.1%
4	83.3%	China	Asia	8	2.1%
5	81.8%	India	Asia	9	2.0%
5	81.8%	Netherlands	Europe	9	2.0%
6	77.8%	France	Europe	10	1.6%
7	66.7%	Israel	Asia	10	1.6%
8	65.0%	Canada	North America	4	3.6%
9	64.7%	Hong Kong	Asia	6	3.0%
10	60.0%	Sweden	Europe	13	0.9%

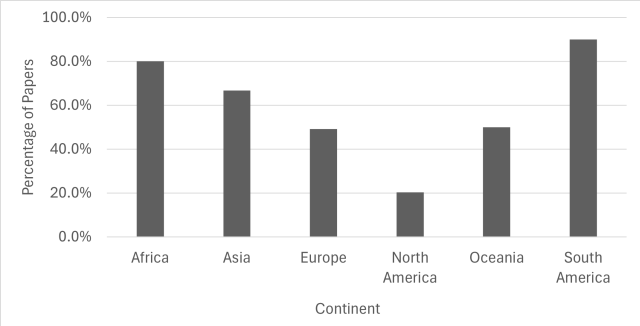

Figure 3: Percentage of collaborative Learning @ Scale papers by continent.

4 DISCUSSION

RQ1: Nearly one-fifth of all Learning @ Scale papers have more than one country represented by their authors. However, this statistic has been trending downwards every year since 2017 (except 2021), and the authorship for the average paper has only one country represented. Are research endeavors becoming more geographically isolated? Is IRC somehow becoming more difficult, in spite of the increasing ease of connectivity and the recent rise in popularity of remote work? Or, from a more optimistic standpoint, is the field of at-scale education research expanding enough to allow authors to more easily find peers to collaborate with in their countries of residence? Undoubtedly there are multiple factors at play; we present these questions as seeds for potential future research.

RQ2: The data suggest that Learning @ Scale authors from South American, African, and Asian countries tend to be more collaborative than authors from other countries. There may well be cultural or political factors influencing this trend, or perhaps there are simpler logistical reasons pushing these authors to collaborate with authors from other countries. Indeed, research has shown that such "extra-scientific factors" can affect IRC [3].

RQ3: Interestingly, although North American and European countries claim the top four spots in terms of countries that have appeared in the authorship of the most papers, countries from these continents claim only three of the top eight spots (top five in the ordering) in terms of the percent of papers they appear on that are collaborative. The other five countries in the top eight are South American and Asian countries. None of the top six countries on the former list appear in the top eight of the latter list. And one of the most glaring trends is that, while the United States is represented among authors on the vast majority of Learning @ Scale papers, the percent of their papers that are collaborative is the lowest of all countries. Even when grouped by continent, the list of continents with the most collaborative papers (as a percentage of all papers coming from authors on that continent) is almost reversed from the list of continents whose authors have appeared on the most papers.

In general, we observe that countries that more frequently appear in the authorships of Learning @ Scale papers tend to produce collaborative papers less frequently than those countries with fewer total papers. It is possible that those countries that publish at Learning @ Scale more frequently do so because they happen to have more residents conducting research in this field (possibly due to simply having a higher total population). If this is true, it may be logistically easier for authors in these countries to collaborate with each other rather than with authors in other countries.

5 LIMITATIONS

While our authorship analysis was comprehensive, representation on Learning @ Scale papers is dominated by a few countries, in particular the United States, and the limited data for less-represented countries is an asterisk on any conclusions we might draw about their representation. More generally, while we discussed a number of findings related to Learning @ Scale authorship, we recognize the complexities inherent in determining causality, as there may be political or cultural factors outside the scope of our research.

6 FUTURE WORK

Based on our metrics, IRC at Learning @ Scale is infrequent and appears to be decreasing over time. One area for future research is exploring the factors driving this trend, which may provide insights into how to improve it.

One glaring finding from our analysis was that the countries frequently represented among Learning @ Scale authors tend to be less represented on collaborative papers, and vice versa. As mentioned, there may be a variety of reasons––historical, cultural, political––for this trend. A closer examination of the causes of this phenomenon (perhaps in collaboration with researchers from different fields) may provide insights into how to increase representation and IRC on Learning @ Scale publications.

7 CONCLUSION

While IRC certainly occurs at Learning @ Scale, there is clear room for improvement, especially among those countries that publish most often at the conference. We encourage Learning @ Scale authors to consider the benefits of IRC and to seek opportunities to collaborate with researchers in other countries and continents. The corpus of existing Learning @ Scale research provides an excellent starting point for discovering researchers, institutions, and organizations around the world with similar research interests.

We also assert that there is an onus on the Learning @ Scale conference to encourage and facilitate IRC. This may involve organizing deliberate efforts to foster international connections, perhaps through dedicated conference sessions, or through less formal networking or social events, or both. Additionally, devoting time during the conference to spotlight some of that year's international collaborations may be beneficial. Further, an award that recognizes one paper each year that best exemplifies "global perspectives in learning at scale through international collaboration" would signal—to Learning @ Scale authors and the broader research community—the conference's commitment to advancing IRC.

International research collaboration benefits researchers and the field of research alike, and it is crucial for obtaining a deeper understanding of global perspectives in research. As the concept of "learning at scale" increasingly implies "global learning", accounting for global perspectives in this field is becoming less of an option and more of a requirement. IRC at Learning @ Scale can provide insights into the generalizability of research findings and help ensure that the research keeps pace with the changing nature of the field itself.

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